Wheel Strategy ETF Fund

Complete System Report - 2025 Backtest

Cornell University - CHEME-5660 Final Project

Framework: Julia | Pricing: CRR American Binomial Lattice

Initial NAV: $600,000,000 | Universe: 35 Names

Report Generated: March 2026

1. System Architecture

The system consists of 8 Julia source modules + 2 entry-point scripts, loaded in dependency order by Include.jl.

File	Role
Include.jl	Environment setup: activate project, load packages, include modules
src/Files.jl	Data loaders: JLD2 market data, SAGBM CSV, Finviz CSV
src/DataDownload.jl	Yahoo Finance API: prices & dividends with CSV caching
src/Compute.jl	Log-growth matrix, rolling vol, div yield, IV calibration, correlation
src/OptionPricing.jl	CRR American option pricing, Greeks, strike_from_delta, IV solver
src/EarningsCalendar.jl	Earnings date mgmt: CSV/estimation/Yahoo, near_earnings detection
src/MonteCarloSim.jl	GBM, regime-switching GBM, correlated GBM, HMM, stress testing
src/WheelEngine.jl	Core backtest engine: state machine, roll, repair, risk overlays, NAV
Backtest.jl	BACKTEST ENTRY: universe -> data -> simulation -> reporting
Wheel Strategy Code	LIVE TEMPLATE: daily signal generation for real-time ops

2. Data Pipeline

Data Sources

Source	Description	Loader
JLD2 Market Data	S&P 500 2025 daily OHLC (Polygon.io)	Files.jl -> MyTestingMarketDataSet()
SAGBM CSV	Pre-computed drift/vol per ticker (2014-2024)	Files.jl -> load_sagbm_parameters()
Finviz CSV	Div yield, market cap, volume	Files.jl -> load_finviz_screener()
Yahoo Finance Prices	Daily OHLC + adj_close	DataDownload.jl -> download_price_data()
Yahoo Finance Divs	Ex-dividend dates + amounts	DataDownload.jl -> download_dividends()
VIX	VIX index daily (reference)	DataDownload.jl
SPY	S&P 500 ETF for benchmark	DataDownload.jl

3. Universe Construction (Varner PDF Sec.3)

Two-Sleeve Portfolio

Sleeve	# Names	Allocation	Per-Name Wt	Selection Criteria
Safe	25	60%	2.4%	Low vol, high div yield (>=3%), deep options
Aggressive	10	40%	4.0%	Top-quintile vol, deep options, sector diversity

Initial NAV: $600,000,000 | Per-name cap: <=5% NAV | Sector cap: <=25%

Two-Block Inventory per Ticker

Block	Allocation	Purpose
Block A (Hold)	50%	Buy-and-hold for dividends + capital appreciation. No options.
Block B (Wheel)	50%	Wheel cycle: Sell Put->Assigned->Sell Call->Called Away

Safe Tickers (25)

PEP, KO, PG, JNJ, CME, CMCSA, VZ, T, IBM, MO, PM, MDLZ, EXC, KMB, PAYX, TROW, PFG, SO, DUK, ED, LNT, GIS, CAG, REG, CPB

Aggressive Tickers (10)

TSLA, NVDA, AMD, AAPL, MSFT, AMZN, GOOG, META, NFLX, DVN

4. Preprocessing Calculations

4.1 Log Growth Matrix (CHEME-5660 Week 5b)

```
mu_{t,t-1}(r_f) = (1/dt) * ln(S_t / S_{t-1}) - r_f
where dt = 1/252, r_f = risk-free rate
```

Computes excess log returns for all tickers from JLD2 dataset. Output: (days x tickers) matrix for cross-sectional volatility estimation.

4.2 30-Day Rolling Realized Volatility

SELF-DESIGNED - no direct course reference

```
sigma_RV(t) = std(r_{t-29}, ..., r_t) * sqrt(252)
```

Annualized rolling volatility using trailing 30-day window of daily log returns. Floor at 1%.

4.3 Dividend Yield

```
q = (sum of dividends) / (avg price) / lookback_years
```

Continuous dividend yield passed to CRR lattice for accurate American option valuation around ex-dividend dates.

4.4 IV Calibration (VRP Model)

SELF-DESIGNED - Entire IV calibration module is original work.
Academic motivation: Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009).
Our specific parametric model (VRP x term x skew x volvol) is original.

```
sigma_IV = sigma_RV x VRP x (1 + term_adj) x (1 + skew_adj) x (1 + volvol_adj)

VRP multiplier:  Safe = 1.15x,  Aggressive = 1.25x
term_adj  = term_slope * max(0, (30/365 - T) / (30/365))
skew_adj  = skew_slope * max(0, -moneyness)
volvol_adj = vol_of_vol * max(0, sigma_RV - 0.30)
```

Data needed for better calibration: (1) Real options chain bid/ask -> extract actual IV; (2) VIX historical data -> time-varying VRP; (3) CBOE DataShop / OptionMetrics for IV surfaces.

4.5 Return Correlation Matrix (CHEME-5660 Week 6)

```
rho_{ij} = cor(r_i, r_j) on pairwise-complete observations
Z_corr = L * Z_indep,  L = cholesky(rho).L
```

5. Option Pricing Model

5.1 CRR Binomial Lattice - American (CHEME-5660 Week 10)

```
dt = T / N
u = exp(sigma * sqrt(dt)),  d = 1/u
p = (exp((r-q)*dt) - d) / (u - d)    risk-neutral prob with div yield q

Backward induction (American):
V_j = max(intrinsic_j,  exp(-r*dt) * [p*V_{j+1} + (1-p)*V_j])
```

N = 50 steps. All pricing uses American exercise exclusively. European pricing is never used because the Wheel trades American-style equity options.

5.2 Greeks (CHEME-5660 Weeks 11-12)

Greek	Method	Formula
Delta	CRR first level	D = (V_u - V_d) / (S*u - S*d)
Gamma	Central finite diff	G = (V(S+e) - 2V(S) + V(S-e)) / e^2
Theta	Forward diff	Th = V(T - 1/365) - V(T)
Vega	Central finite diff	v = (V(s+0.01) - V(s-0.01)) / 0.02

5.3 strike_from_delta (SELF-DESIGNED)

SELF-DESIGNED - no course or textbook reference

Input: target |Delta| (e.g. 0.25), S, sigma, T. Method: Binary search over K in [0.5S, 2.0S], compute CRR delta at each midpoint, converge when K_hi - K_lo < $0.005.

5.4 estimate_implied_vol (SELF-DESIGNED)

SELF-DESIGNED - CRR-based alternative to BSM IV solver

Input: market option price, S, K, T. Method: Binary search over sigma in [0.01, 3.0], price via CRR at each midpoint, converge when sigma_hi - sigma_lo < 1e-6.

6. Backtest Engine

6.1 State Machine per Ticker

```
SELLING_PUT -> (put assigned: S < K) -> HOLDING_SHARES -> (call ITM: S > K) -> SELLING_PUT
```

Each ticker's Block B has 1-3 ladder slots, each running an independent Wheel cycle for temporal diversification of expiry dates.

6.2 Daily Simulation Loop

Step	Action	Ref
1	Deduct daily mgmt fee (0.68% annual / 252)	PDF S5
2	Compute trailing VaR/ES -> throttle if exceeded	PDF S4,S6
3	Compute drawdown -> feed into adaptive delta	PDF S3
4	Per ticker x slot: deduct borrow cost (0.5% ann)	PDF S7A
5	Check dividends -> credit Block A + B held shares	PDF S2
6	Check option expiry -> assignment or call-away	PDF S2
7	Check 4 roll triggers -> roll if triggered	PDF S3
8	If no option: check risk/earn/caps -> open_option!	PDF S3-6
9	Check cost-basis repair (loss >=10% -> avg down)	PDF S6
10	Record DailyRecord: NAV, Greeks, cumulative totals	-

6.3 Roll Triggers (4 Conditions)

1. Time-based: DTE in [3,5] and OTM
2. Premium decay: current value / original < 20% (80% captured)
3. Moneyness band: |moneyness| > 5%
4. Breakeven breach: price beyond BE +/- 2%

6.4 Trading Cost Model

Component	Amount
Commission	$0.65 / contract
Exchange fee	$0.03 / contract
Clearing fee	$0.02 / contract
Bid-ask slippage	Volatility-adjusted spread model
Borrow fee	0.5% annual on held shares
Management fee	0.68% annual on NAV

6.5 Earnings Policy

Policy	Behavior
:avoid	Skip opening options within +/-5 days of earnings
:widen	Open but reduce delta (more OTM)
:reduce_size	Open but reduce contracts to 50%

7. Monte Carlo & Stress Testing

7.1 GBM (CHEME-5660 Week 5b)

```
S_{t+dt} = S_t * exp[(mu - sigma^2/2)*dt + sigma*sqrt(dt)*Z],  Z ~ N(0,1)
```

7.2 Correlated Multi-Asset GBM (CHEME-5660 Week 6)

```
Z_corr = L * Z_indep,  L = cholesky(rho).L
Each asset j: S_j(t+dt) = S_j(t) * exp[(mu_j - sigma_j^2/2)*dt + sigma_j*sqrt(dt)*Z_corr_j]
```

7.3 Regime-Switching GBM (SELF-DESIGNED)

2-state Markov chain switching between normal (mu1, sigma1) and stressed (mu2, sigma2) regimes with transition probabilities p12 and p21. Motivated by Varner PDF Section 7B.

7.4 HMM 2-State (CHEME-5660 Week 13)

```
EM (Baum-Welch): Forward (alpha) + Backward (beta) -> gamma, xi
M-step: Update mu_k, sigma^2_k, transition matrix A
Output: (mu_normal, sigma_normal, mu_stressed, sigma_stressed, p12, p21)
```

7.5 Stress Scenarios

Scenario	Vol Mult	Drift Adj	Gap
Normal	1.0x	0	-
Vol Spike	2.0x	0	-
Bear Market	1.5x	-20%	-
Flash Crash	2.0x	0	-10% d30
Name Blowup	1.5x	0	-30% d60
Bull Squeeze	1.5x	+30%	+15% d45

8. Results & Charts

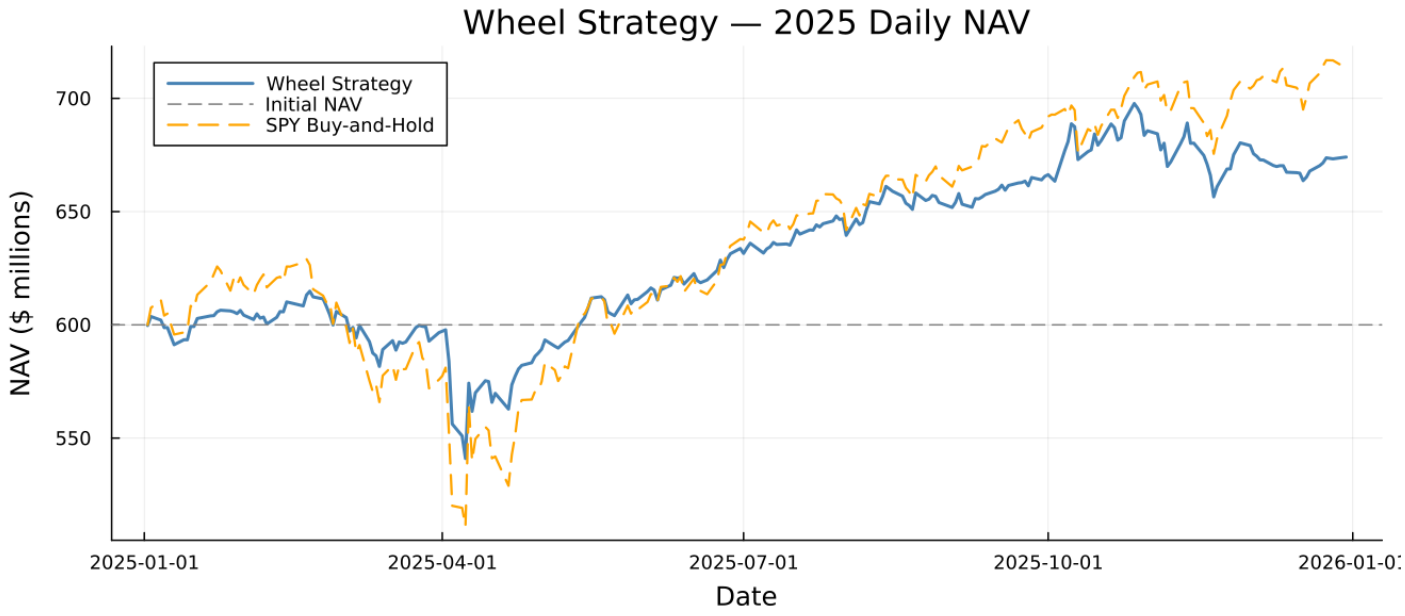

Chart 1: NAV Curve vs SPY Benchmark

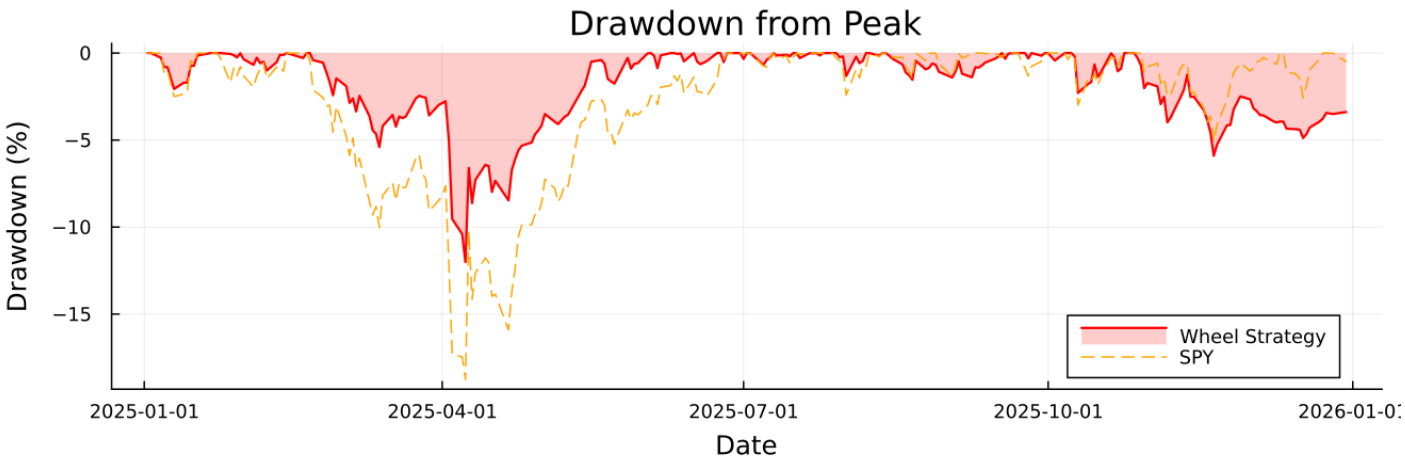

Chart 2: Drawdown from Peak (Wheel vs SPY)

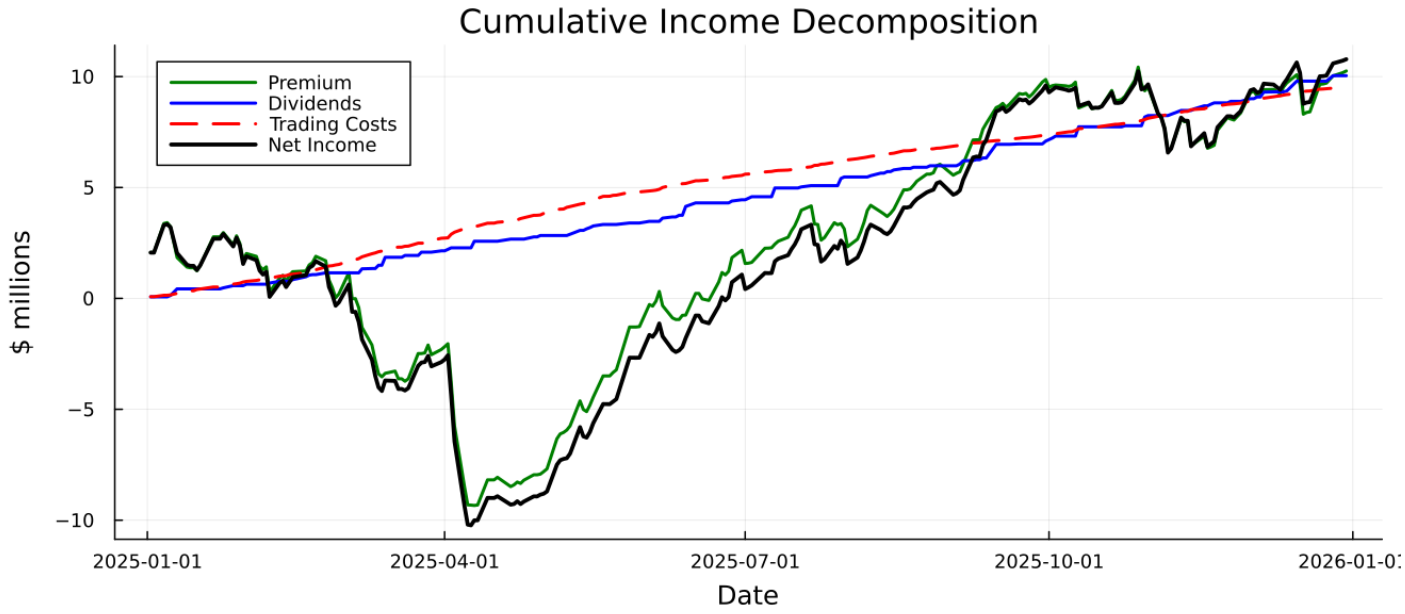

Chart 3: Cumulative Income Decomposition (Premium + Divs - Costs)

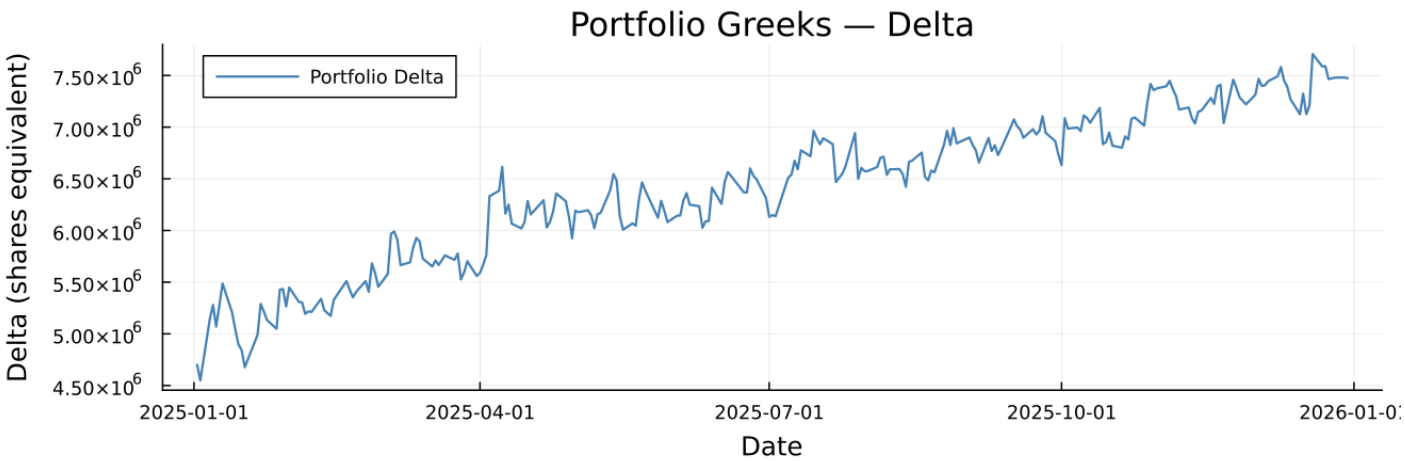

Chart 4: Portfolio Delta over Time

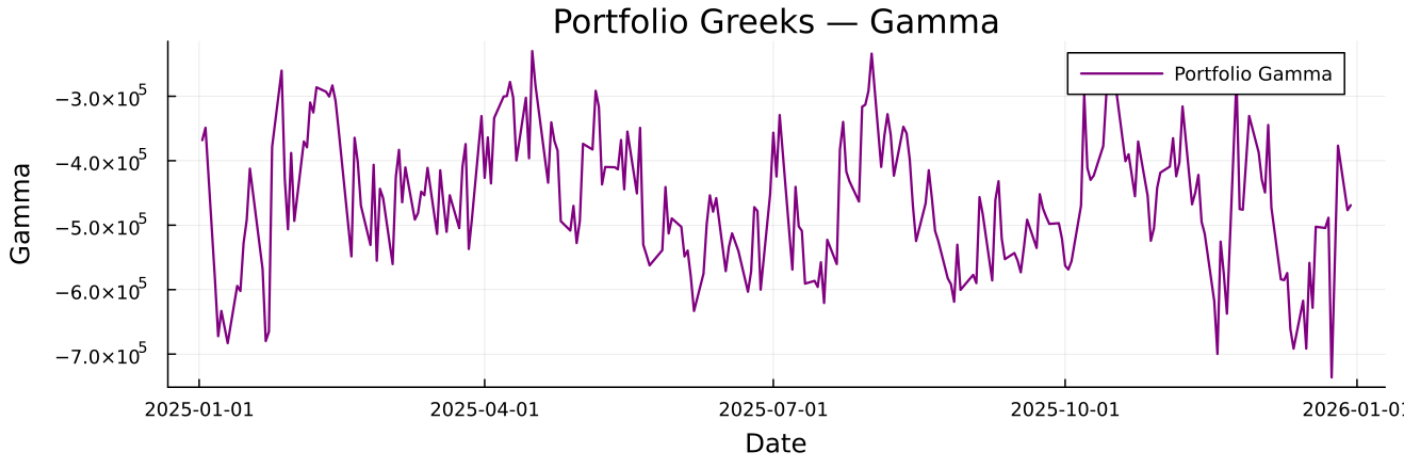

Chart 5: Portfolio Gamma over Time

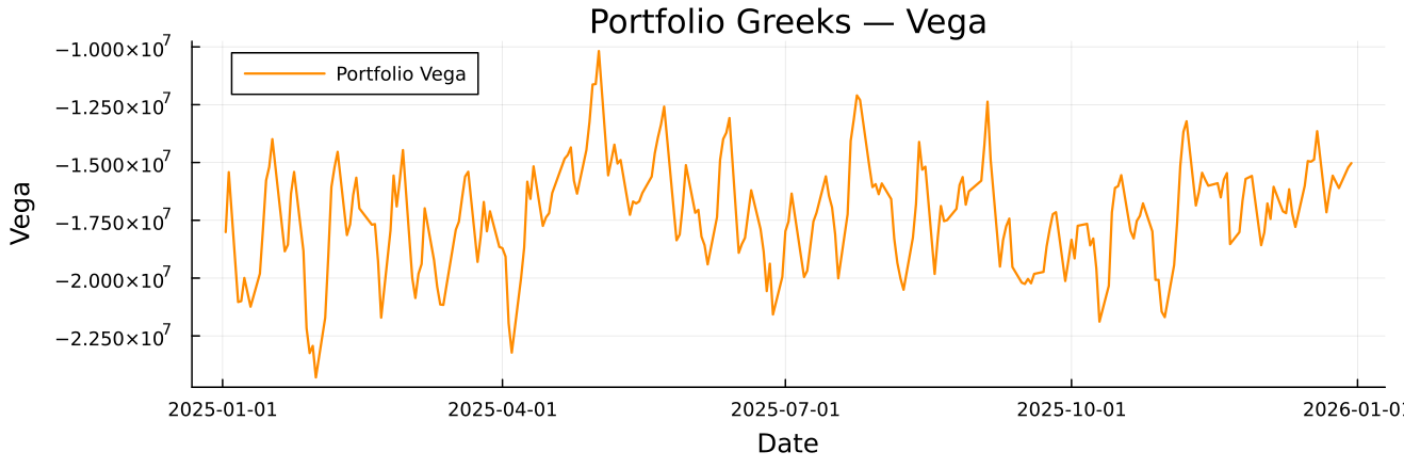

Chart 6: Portfolio Vega over Time

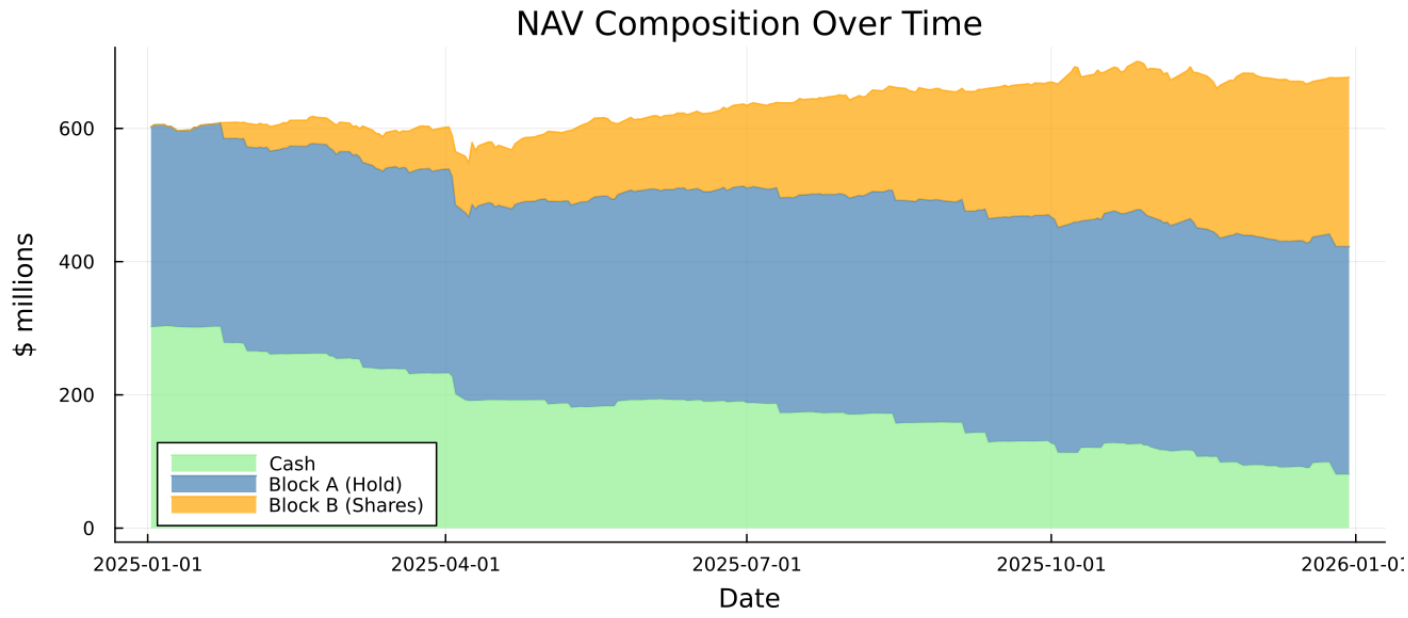

Chart 7: NAV Composition (Cash / Block A / Block B)

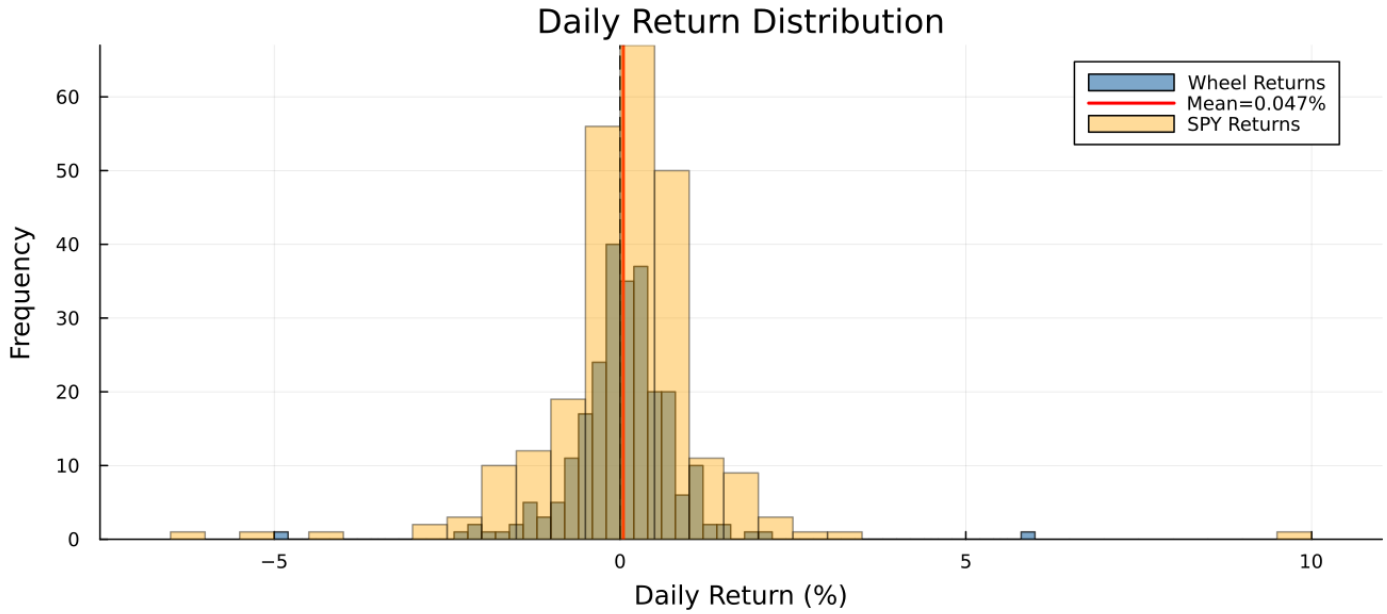

Chart 8: Daily Return Distribution (Wheel vs SPY)

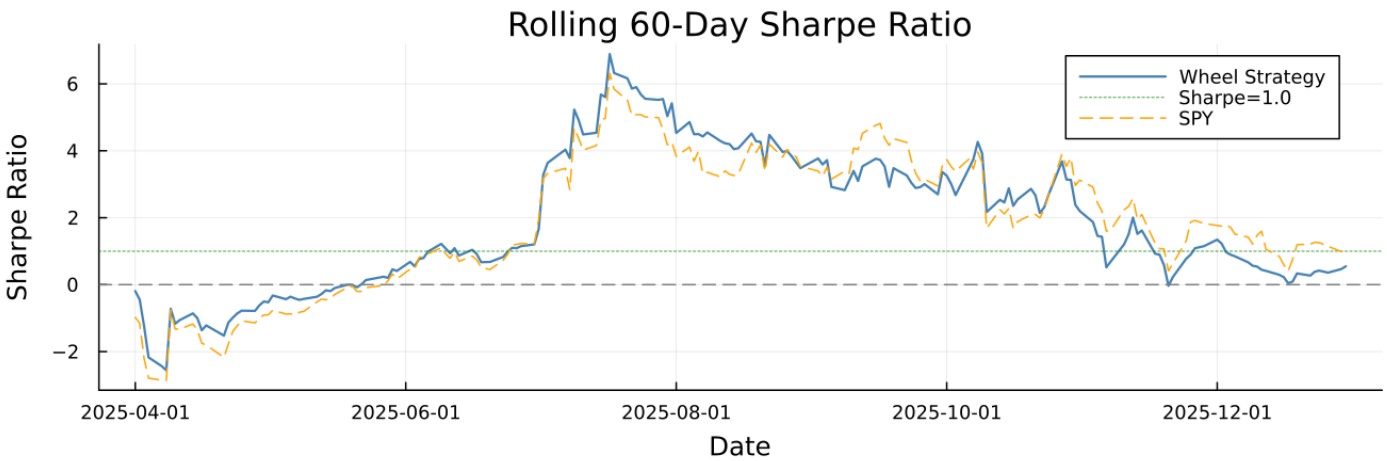

Chart 9: Rolling 60-Day Sharpe Ratio

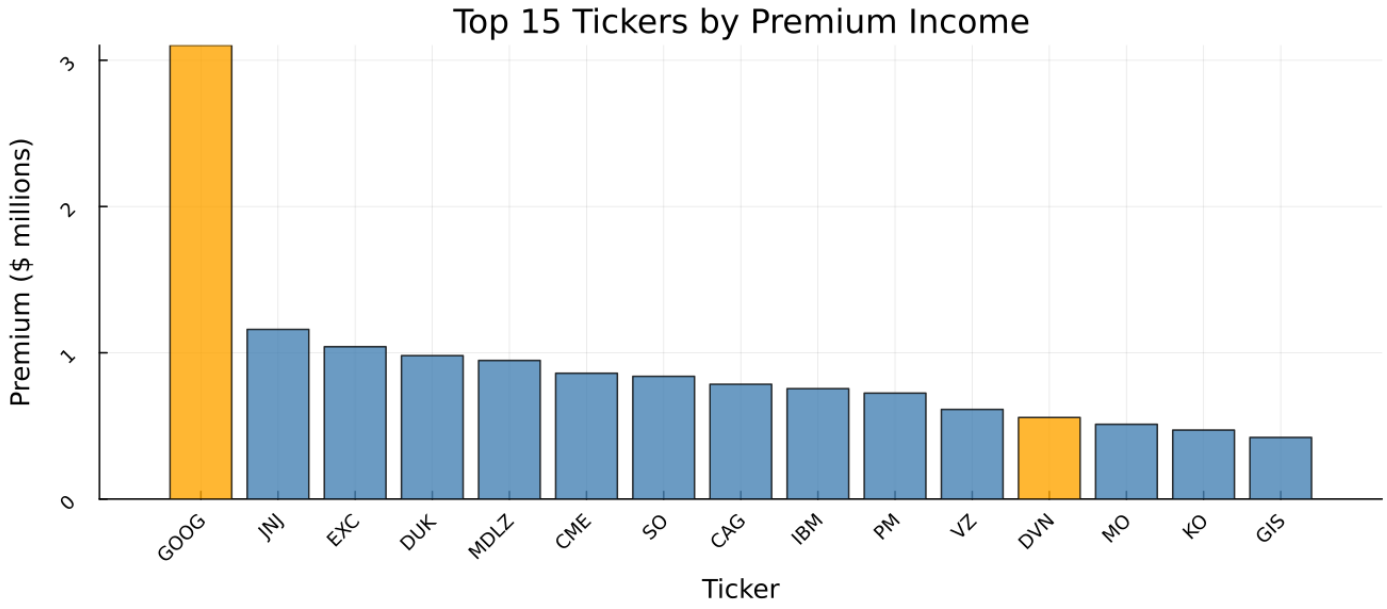

Chart 10: Top 15 Tickers by Premium Income

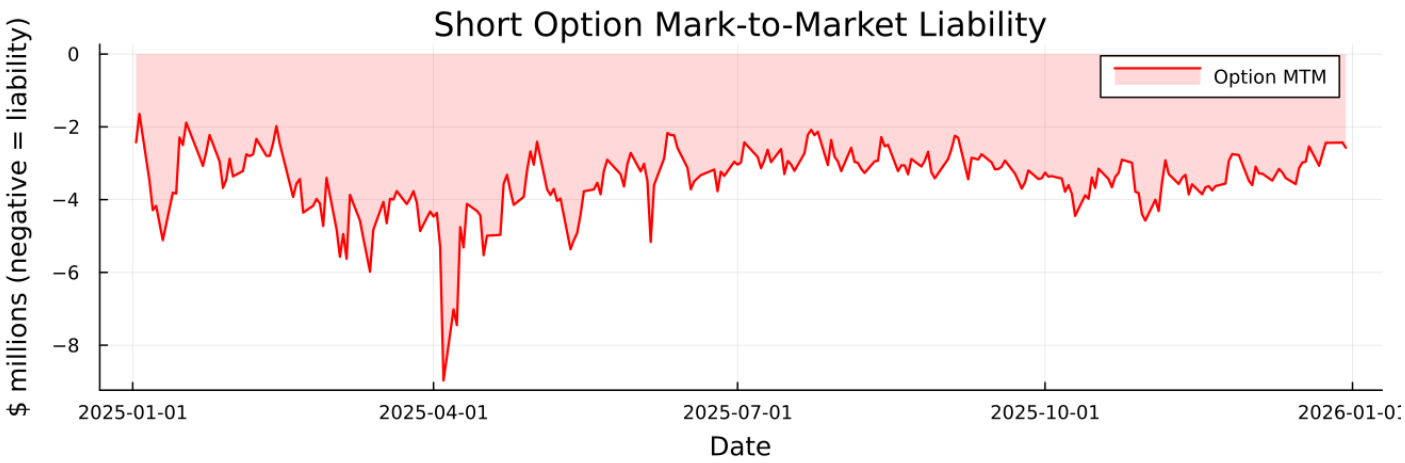

Chart 11: Short Option Mark-to-Market Liability

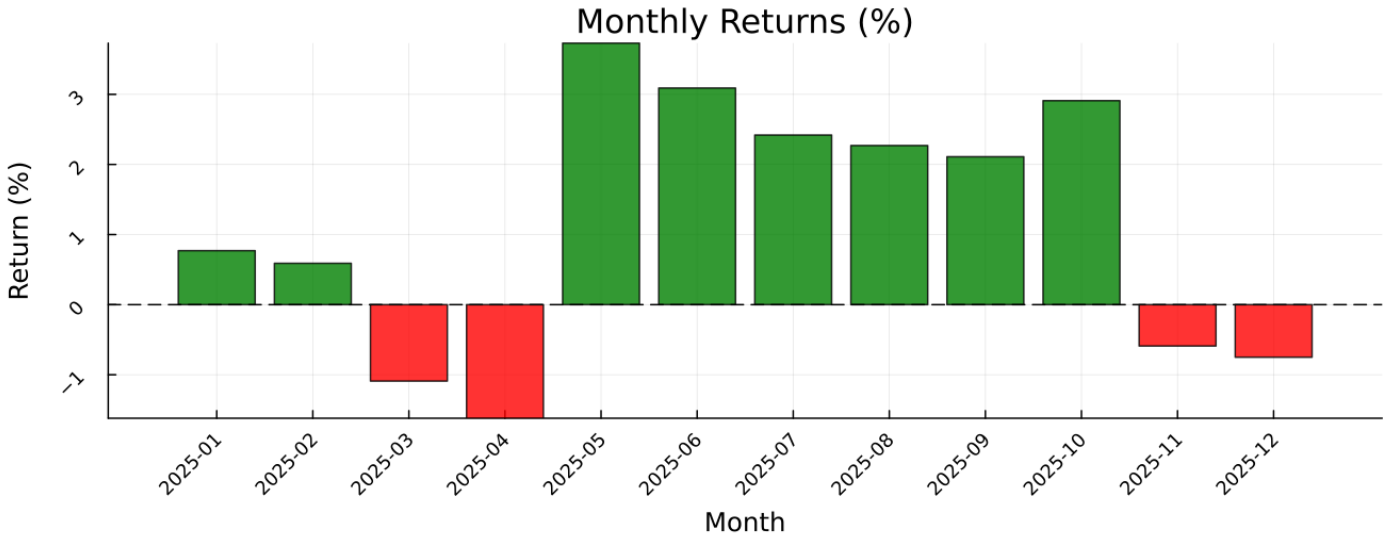

Chart 12: Monthly Returns

9. Output Files

CSV Data Files

File	Key Columns	Purpose
daily_nav_2025.csv	Date,NAV,Cash,Delta,Gamma,Vega,DailyReturn,SPY...	Day-level full dataset for downstream analysis
ticker_performance_2025.csv	Ticker,Premium,Dividends,Costs,Assigns,Trades...	Per-ticker attribution and performance

Chart Files (12 PNGs)

#	File	Analysis Purpose
1	nav_curve_2025.png	Core performance: did strategy beat benchmark?
2	drawdown_2025.png	Risk: max drawdown depth & recovery
3	income_decomposition_2025.png	Income source decomposition, Distribution Yield
4-6	greeks_*.png	Directional, convexity, volatility risk monitoring
7	nav_composition_2025.png	Asset allocation visualization over time
8	return_distribution_2025.png	Return profile characterization
9	rolling_sharpe_2025.png	Time-varying risk-adjusted performance
10	premium_by_ticker_2025.png	Per-name contribution attribution
11	option_mtm_2025.png	Outstanding short option exposure
12	monthly_returns_2025.png	Monthly performance review

10. Course Reference Mapping

CHEME-5660 Content Implemented

Week	Topic	Implementation
5b	GBM / SAGBM parameter estimation	log_growth_matrix(), simulate_gbm()
6	Multi-asset GBM + Cholesky	simulate_correlated_gbm(), compute_return_correlation()
10	CRR Binomial - American Options	crr_price() with dividend yield q
11	Greeks (D, G, Th, v)	crr_delta/gamma/theta/vega(), option_greeks()
12b	Delta and hedging	crr_delta() -> strike_from_delta()
13	HMM - Markov Models	fit_two_state_hmm(), classify_regime()

Self-Designed Components (not from course)

Component	Description
IV Calibration (VRP)	sigma_RV->sigma_IV via Variance Risk Premium+term+skew+volvol
strike_from_delta	Bisection search: find K producing target |Delta| under CRR
estimate_implied_vol	Bisection on CRR to find sigma matching market price
Bid-ask spread model	Volatility-adjusted half-spread estimation
Fill rate model	Spread-based fill rate with vol penalty
Regime-switching GBM	2-state Markov x GBM for regime-aware simulation
Earnings calendar	SEC filing pattern-based quarterly date estimation
Cost-basis repair	Automated average-down when shares >=10% underwater
Adaptive tenor/delta	Vol-regime-aware selection of tenor and delta target
Rolling IV computation	Convert time-series sigma_RV to sigma_IV for all tickers

11. Improvements & Future Work

Priority Improvements

Priority	Improvement	Current State	Method
HIGH	IS/OOS + Walk-Forward	Only in-sample 2025	2014-2024 train, 2025 OOS. PDF S7A required.
HIGH	Integrate HMM to backtest	HMM done, not integrated	Daily classify_regime -> adjust delta/tenor
HIGH	Real IV Data	Fixed VRP multipliers	CBOE/OptionMetrics/broker API for IV surface
MED	Correlated Stress Test	Per-ticker / uniform	simulate_correlated_gbm + correlations
MED	Execution Realism	Close-price execution	Open/VWAP/next-day-open fill simulation
MED	Dynamic Rebalance	Weights fixed	Monthly/quarterly rebalance Block A/B
LOW	Heston Model	Rolling realized vol	Replace GBM with stochastic vol
LOW	TWAP/VWAP Execution	Fixed slippage	Real L2 data + optimal execution
LOW	Live Broker API	Template only	Interactive Brokers TWS API (IBApi.jl)

Varner PDF Requirements vs Implementation

Section	Requirement	Status
S2	Wheel cycle: Short Put -> Assigned -> CC -> Called Away	DONE
S3	Safe/Aggressive, 5% name cap, sector cap, laddering	DONE
S4	Distribution Yield, Premium Capture, Sharpe, VaR/ES	DONE
S5	Commissions, exchange fees, borrow fees	DONE
S6	Delta range, tenor, laddering, earnings, roll, repair	DONE
S7A	IS/OOS split, walk-forward	NOT DONE
S7A	Parameter sweep	DONE (flag)
S7B	Regime-switching GBM, gap, vol spike	DONE (flag)
S7B	IV surface calibration	PARTIAL (VRP model)